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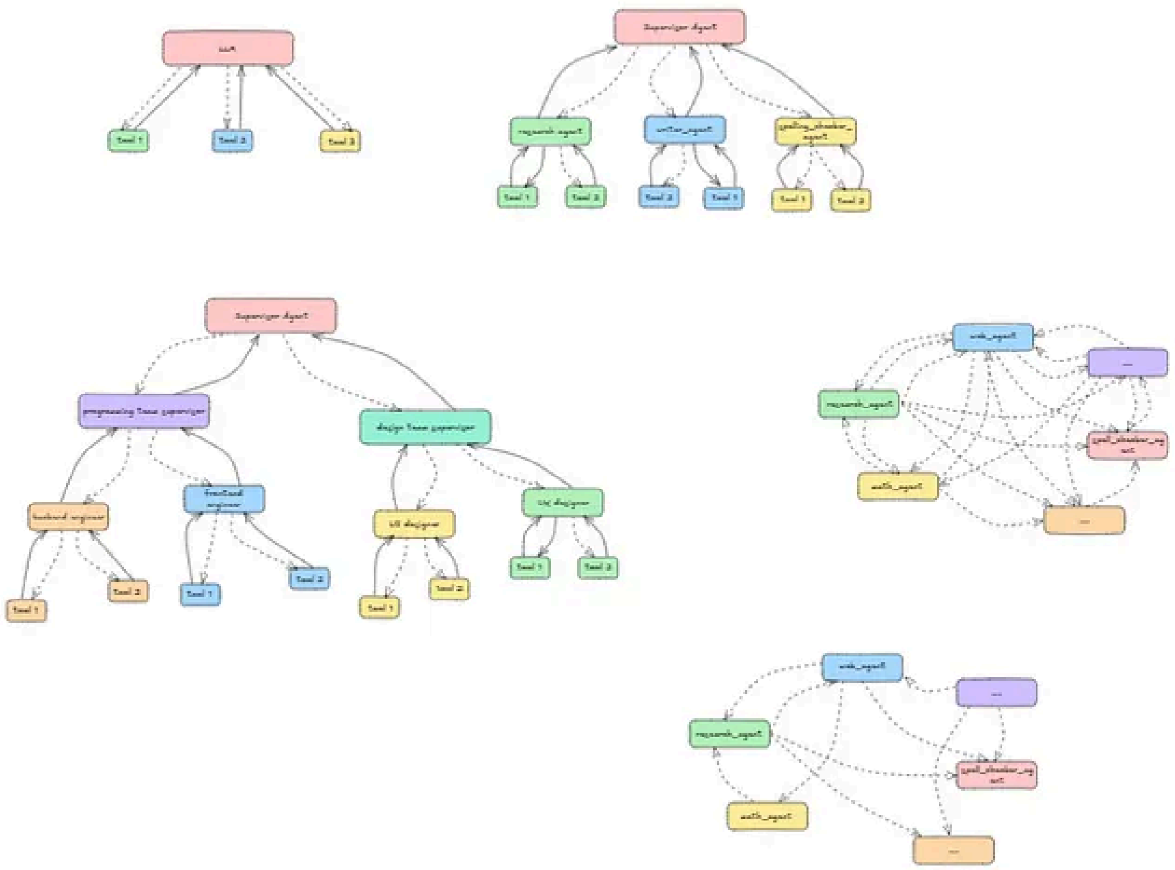
Hello guys, welcome back to another exciting article on working with multi-agent AI systems. In this article, we'll dive into the basics of multi-agent architecture, we'll go over a couple of multi-agent architectures mainly:

1. Single Agents
2. Supervision
3. Hierarchical
4. Network
5. Custom
6. Swarm Agent Architecture



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Background

Have you ever been in a place where you have built a multi-agent system with few agents and few tools? As requirements grow bigger and bigger, more and more tools are coming into your systems, and agents have multiple tools, sometimes as many as 10 tools.

This can lead agents to be confused on what tool to use; agents begin to make poor decisions on tool selection. At this point, it can really get frustrating. One quick solution is to look into your code and try to understand what the underlying architecture of your program is. This can only help if you truly understand your agent architecture and what the agents are to do, what tasks your agents are designed to do.

In this article, we'll go over a basic understanding of these multi-agent architectures and in the upcoming articles, we'll dive deep into each one specifically.

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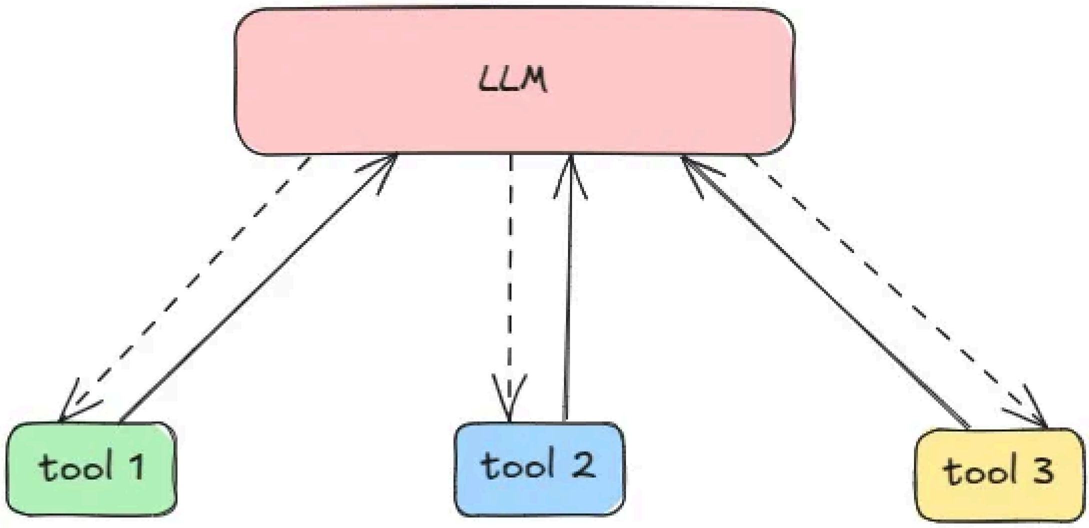
Single Agent Architecture

In a single agent architecture, this is the simplest to understand. In this approach, we only have one agent with a host of tools and connected to an LLM. The agent then takes in a query and makes decisions on what tool to call using the LLM for reasoning.



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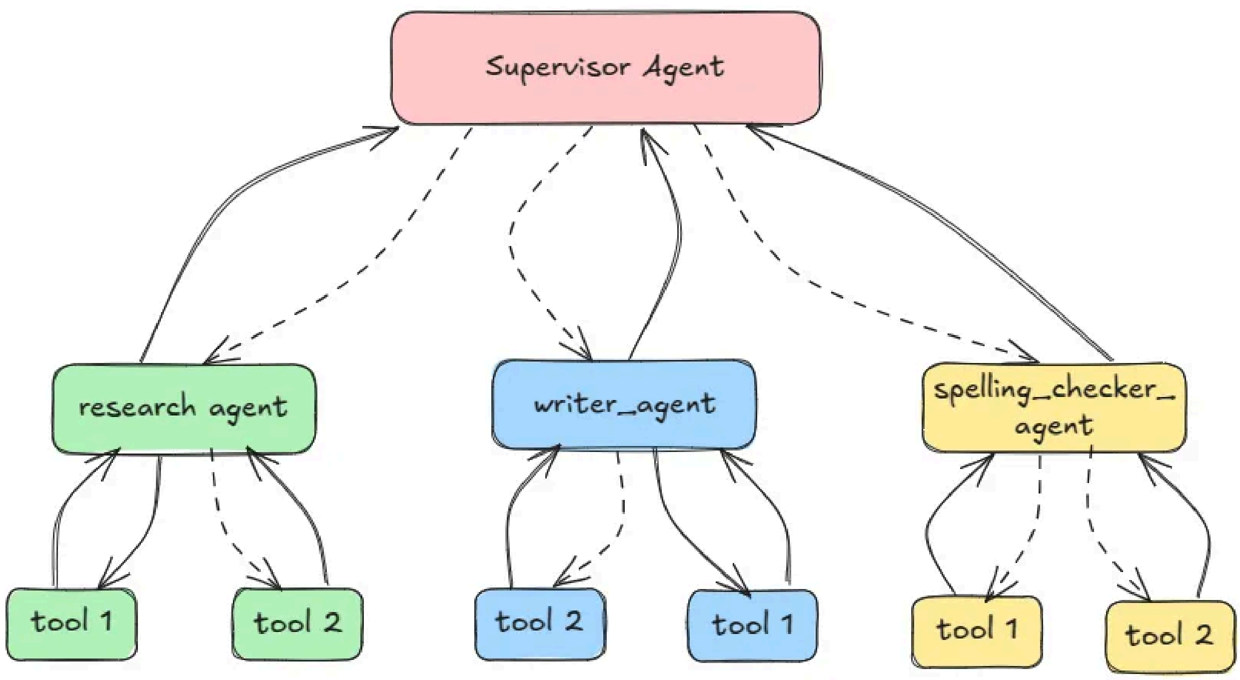
Supervisor Agent Architecture

In the supervisor architecture, we have one agent that has multiple other agents (sub-agents) under it. The supervisor, basically, you can think of it as the orchestrator in a church choir. It takes in queries, uses an LLM to reason over it and decide on what agents to delegate a task to or a sub-task to. It receives the input from these agents, reasons over it and decides whether to call another agent if needed or not, if not, it returns the final output.



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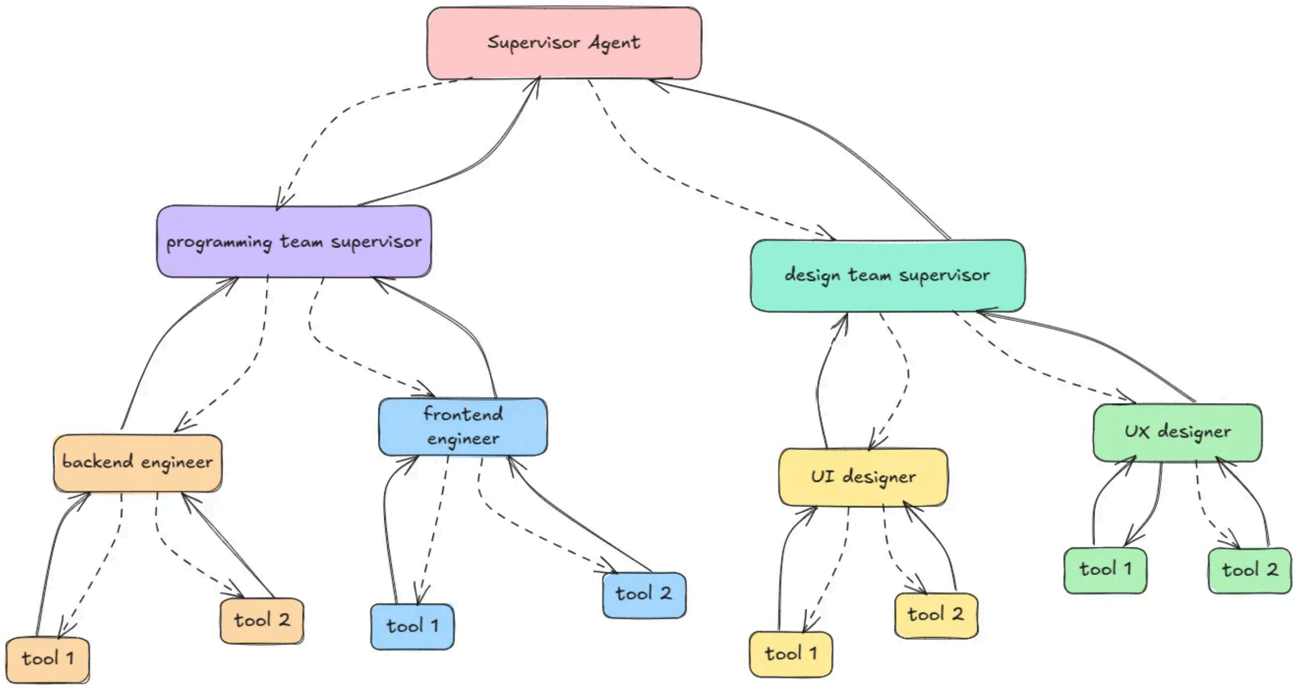
Hierarchical Agent Architecture

Sometimes, having only one supervisor that makes all the decisions and deciding on what agent to call, the agent can start to under-perform due to so many things it has to handle and be an expert at handling. In such cases, one common approach is create teams, where each team has its own supervisor, that supervisor is specialized in managing its own team of sub-agents. Basically breaking the graph into subgraphs, each sub-graph has its supervisor and sub-agents under that supervisor.



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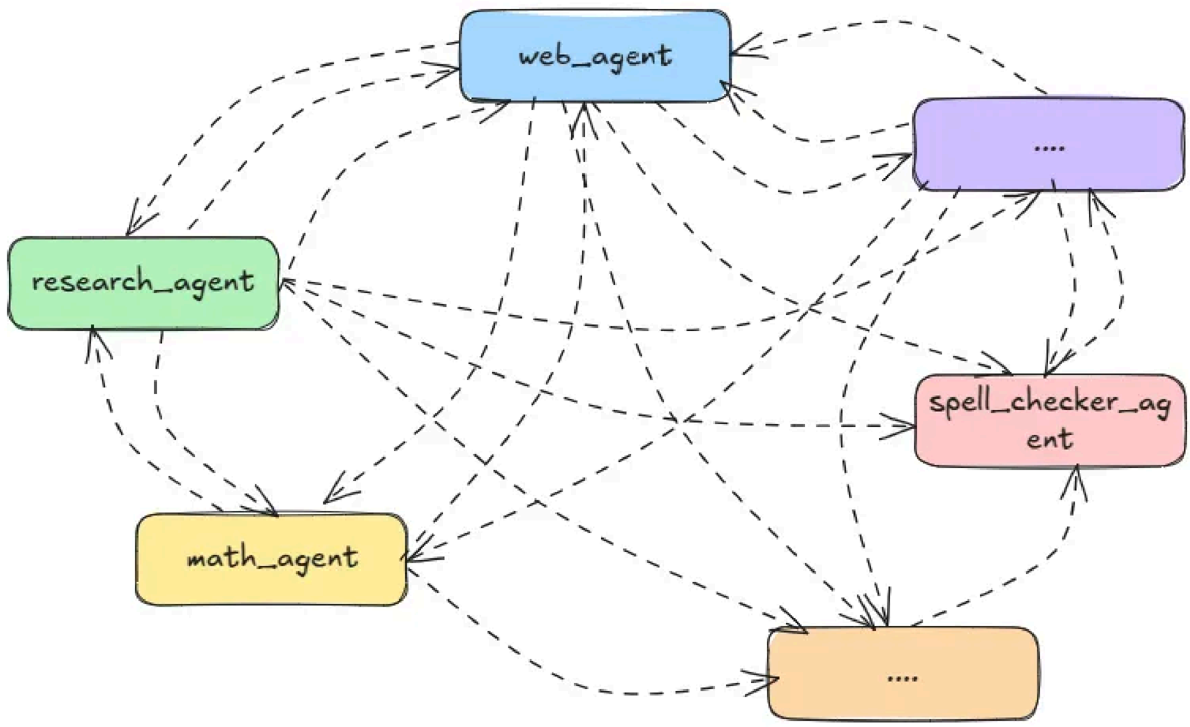
Network Agent Architecture

Networking, I personally understand this with experiences from normal day life. In a normal life setting, you can talk to anyone and they can give you feedback. You can shoot a message to a person you admire on LinkedIn, or any girl/boy you like right? Same concept here, but now with agents, each agent can talk to any other agent in the whole collection.



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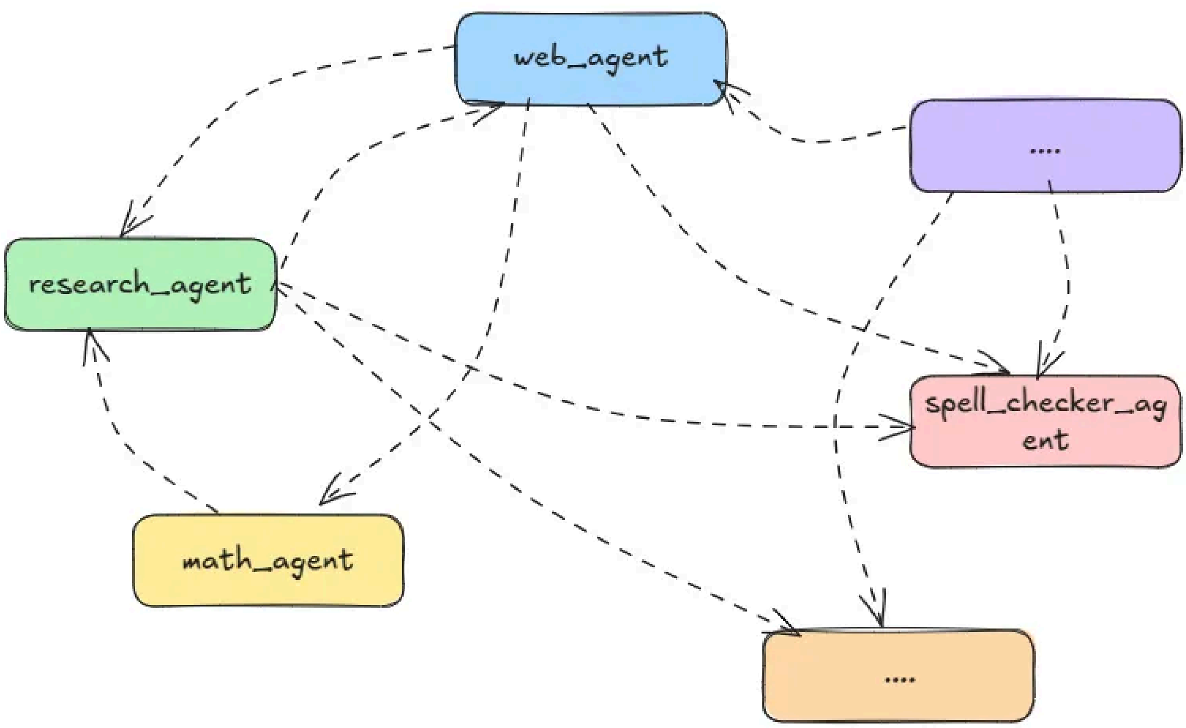
Custom Agent Architecture

In this architecture of agents, each agent can only communicate with selected sub agents and unlike networking where each agent can communicate with all other agents.



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Swarm Agent Architecture

A swarm agent architecture is a type of multi-agent system where specialized agents dynamically pass control to one another based on their areas of expertise, with the system maintaining memory of which agent was last active to ensure conversation continuity.

Conclusion

Congratulations for making it this far, in this article, we went over the theoretical understanding of multi-agent architectures. Powered with this knowledge, let's now dive into each individual project and understand the implementation from the code standpoint.

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Happy coding! And see you next time, the world keeps spinning.

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Hello there , am Prince a full-stack web developer, data science enthusiast, lover of Python Programming with a deep interest in deep learning, computer vision